

Adult Prediction of Children's Perceived Media Pacing

Study Aims and Stimulus Criteria

Working protocol for Samuel and Mia - Cedarville University undergraduate psychology study

Current design: Option 3.5, Replicated Feature Design

Status: Study-development document; not a substitute for faculty, methods, IRB, or statistical review

The study in one sentence

We are studying whether adults can predict how children perceive the pace of short *Curious George* clips, and whether adult-child agreement looks different when clips are selected to contrast primarily in cut rate, visual motion, or audio intensity.

Why this question matters

Adults routinely make judgments about how media will feel to children, but an adult's prediction may be anchored in the adult's own experience. The study therefore separates three quantities:

1. how fast an adult says a clip feels to the adult;
2. how fast that adult predicts the clip would feel to a child;
3. how fast children say the same clip feels to them.

CMAT supplies objective descriptions of the stimulus. This lets us ask whether adult predictions and child ratings correspond to measured formal features, rather than reducing agreement or disagreement to one unexplained number.

The study is about **perceived pace**. It is not a test of whether a program is overstimulating, harmful, healthy, educational, or developmentally appropriate.

Research questions

Primary question

Can adults accurately predict children's reported perceptions of pacing across selected *Curious George* clips?

Secondary question

Does adult-child agreement differ across matched clip contrasts selected to vary primarily in:

- cuts per minute;
- visual motion; or
- audio intensity?

Exploratory questions

- Are adult predictions strongly anchored in adults' own ratings?
- Do two independent contrasts for the same feature show similar rating patterns?
- Do adults and children appear to respond differently to particular formal features when judging pace?
- Do CMAT measurements provide a practical and reproducible way to locate, describe, and freeze stimuli for a human-participant study?

Feature-level interpretation is exploratory. Two pairs per feature cannot establish the independent or general importance of cuts, motion, or sound.

Planned participants and response task

The working child age range is **8 to 12 years** because children in this range are more likely than younger children to use the same five-point rating scale meaningfully. If recruitment must be narrowed, the lower end of the band may better balance scale usability with proximity to the program's intended audience.

After every 30-second clip, participants answer:

How fast did this video feel?

The response scale is:

1. Very slow
2. Slow
3. In between
4. Fast
5. Very fast

Children complete one own-perception block. Adults complete two separate blocks:

- **Adult self:** how fast the clip felt to the adult;
- **Adult prediction:** how fast the adult thinks it would feel to a child of the study's specified target age.

The adult target age must be one fixed, clearly worded age that matches the children actually recruited. Adult block order and clip order must be counterbalanced or randomized according to a plan frozen before data collection.

Why use *Curious George*

Using one program holds many production characteristics more constant than a cross-program comparison would. The clips share characters, visual style, general production context, and intended format. This improves control while searching for naturally occurring differences in cuts, motion, and sound.

The tradeoff is limited generalizability. Results from one season of one series cannot automatically be generalized to other programs, genres, production styles, or child age groups.

The required stimulus set

The final study uses **12 unique, 30-second clips** arranged into **six matched analytical pairs**. Participants rate every clip individually. They do not make a forced-choice decision between the two clips in a pair; the pairing is an analysis and stimulus-selection structure.

Pair	Lower/comparison clip	Higher/feature clip	Selection objective
CUTS 1	Clip A1 - lower cuts	Clip B1 - higher cuts	Motion and audio intensity as similar as possible
CUTS 2	Clip A2 - lower cuts	Clip B2 - higher cuts	Independent replication with different scenes
MOTION 1	Clip C1-L - lower motion	Clip C1-H - higher motion	Cuts and audio intensity as similar as possible
MOTION 2	Clip C2-L - lower motion	Clip C2-H - higher motion	Independent replication with different scenes
AUDIO 1	Clip D1-L - lower audio intensity	Clip D1-H - higher audio intensity	Cuts and motion as similar as possible
AUDIO 2	Clip D2-L - lower audio intensity	Clip D2-H - higher audio intensity	Independent replication with different scenes

No clip is reused in another pair. There is no common baseline repeatedly shown to participants.

What CMAT is looking for

CMAT divides the complete candidate pool into relative thirds for each primary measure:

- **Low:** bottom third of the measured candidate distribution;
- **Middle:** middle third;
- **High:** top third.

Ties at a boundary remain tied. These labels describe this *Curious George* candidate pool, not an absolute or clinical definition of low or high.

Primary target: matched feature contrasts

For each feature, we want a large low-to-high difference in the target feature and the smallest practical differences in the other two features. CMAT ranks pairs using percentile separation so measures with different units are not treated as if their raw scales were interchangeable.

The matching priorities are:

1. a clear separation in the target feature;
2. close matching on the two control features;
3. two independent pairs for each feature;
4. 12 unique clips with no clip reuse;
5. different source episodes within a pair;
6. normally no more than two selected clips from one episode;
7. scenes that remain defensible after human review.

Secondary target: informative feature profiles

Profiles such as the following are useful for understanding whether the season contains enough variation:

- high cuts / low motion / low audio intensity;
- low cuts / high motion / low audio intensity;
- low cuts / low motion / high audio intensity;
- low cuts / low motion / low audio intensity; and
- high cuts / high motion / high audio intensity.

The rare off-diagonal profiles are especially informative because cuts and motion often rise together.

However, Option 3.5 does **not** require every possible high/low combination. A strong matched contrast can be useful even if one or both clips contain a middle-level control feature.

Why these three measures

Cuts per minute

Cuts describe editing frequency. A clip with more detected shot boundaries per minute may feel faster even when activity within each shot is similar.

Visual motion

Motion describes frame-to-frame visual change within and across the sampled frames. A clip can contain substantial movement even without frequent editing, which is why motion must be considered separately from cuts.

Audio intensity

CMAT currently uses linear RMS. This is an objective signal-amplitude measure, not perceptual loudness and not LUFS. It must be called **audio intensity** in study materials and reporting.

Two clips can share a similar mean RMS while having different dynamics, music, speech, or sound effects. Human review therefore remains necessary.

Candidate-pool rules

- Use contiguous 30-second windows.
- Exclude the same verified opening and closing durations from every episode when those sections are genuinely repeated.
- Do not include the final shorter remainder unless the protocol explicitly changes from 30-second stimuli.
- Preserve absolute source timecodes for review and reproducibility.
- Use one source set consistently. Do not mix HD and SD measurements in one candidate distribution.
- Measure the same files participants will see. If finalist clips are transcoded, re-measure the exported participant-facing files.
- Keep the CMAT measurement configuration frozen while building the candidate pool and selecting stimuli.

The local HD and SD seasons are not identical masters: episode durations, audio gain, and some measured cut counts differ. SD may be faster to process, but SD timecodes and measurements cannot be treated as substitutes for HD.

Human review requirements

CMAT proposes numerical matches; Samuel and Mia make the final stimulus decision. Every proposed clip and plausible alternative should be watched in full.

Reject or replace a clip when it contains:

- an opening, closing, title card, credits, station mark, or repeated bumper;
- a technical defect, damaged audio, unexpected silence, or decoding problem;
- a clip boundary that begins or ends at a confusing or unusually abrupt point;
- content that depends heavily on context outside the 30-second window;
- unusually distinctive dialogue, music, narrative tension, comedy, or visual novelty that makes the proposed pair obviously incomparable;

- substantially repeated footage from another selected clip;
- content that raises participant-welfare or IRB concerns; or
- another salient difference likely to overwhelm the intended feature contrast.

Prefer clips that:

- form a coherent, watchable 30-second segment;
- are understandable without extensive prior context;
- have clean audio and video throughout;
- represent different scenes and preferably different episodes;
- preserve the intended target-feature contrast after export and remeasurement;
- are reasonably similar to their pair partner on obvious narrative and formal properties not measured by CMAT.

Human review does not turn naturally occurring clips into an experiment that manipulates only one feature. It improves the defensibility of the matched comparison while leaving the interpretation associational.

Preliminary feasibility evidence

As of August 17, 2026, the first three HD episodes produced **126 eligible 30-second candidates** after excluding the first 60 seconds and last 90 seconds of each episode.

Preliminary correlations across those windows were:

- cuts with motion: $r = .645$;
- cuts with audio intensity: $r = -.056$;
- motion with audio intensity: $r = .106$.

The pool already contained:

- 1 high-cuts / low-motion / low-audio clip;
- 1 low-cuts / high-motion / low-audio clip;
- 13 low-cuts / low-motion / high-audio clips;
- 11 low-cuts / low-motion / low-audio clips; and
- 8 high-cuts / high-motion / high-audio clips.

When the episode-diversity constraint was temporarily ignored for feasibility checking, the 126-window pool contained enough numerical variation to construct two candidate contrasts for each feature. This is encouraging but not a final selection result. The observations are nested within only three episodes, the pool is incomplete, and none of the proposed scenes has passed final human review.

What would count as a successful final stimulus set

Before building the participant task, the final set should satisfy all of the following:

- 12 unique, exactly identified participant-facing files;
- two defensible matched contrasts for cuts;
- two defensible matched contrasts for motion;
- two defensible matched contrasts for audio intensity;
- no repeated common baseline;
- no clip reuse across pairs;
- acceptable human review of every clip and pair;
- CMAT remeasurement of the exact exported files;
- a frozen manifest containing sources, absolute source timecodes, exclusions, measurements, feature labels, pair assignments, software configuration, and file identities;
- a frozen participant order/counterbalancing plan; and
- faculty/methods and IRB approval of the final stimuli and procedure.

What the study can claim

The study may report:

- how well adults predicted children's aggregate pace ratings for the selected clips;
- whether adults' predictions differed systematically from adults' own ratings;
- associations between the three CMAT measures and ratings within this clip set;
- whether adult-child agreement differed across the six matched contrasts;
- whether similar patterns appeared in both pairs representing a feature; and
- whether CMAT provided a usable, reproducible method for finding and characterizing stimuli for this study.

What the study cannot claim

The study cannot establish:

- that cuts, motion, or audio intensity caused a pacing judgment;
- the independent or general importance of any feature across children's media;
- effects on attention, cognition, arousal, behavior, or development;
- whether a program is overstimulating, harmful, healthy, educational, or appropriate;
- how a particular individual child will respond;
- that findings generalize beyond the selected clips, sampled participants, *Curious George*, or the recruited age range; or

- that one study validates CMAT as a whole, including every measure, detector, recipe, construct, or composite in the software.

Related workflow documentation

Technical instructions for running the candidate-pool measurement, reading its outputs, exporting finalists, and freezing provenance are in [STUDY_CLIP_SELECTION.md](#).